

Defect engineering in Czochralski silicon through tin-doping: A perspective from tin-vacancy-oxygen interactions

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ABSTRACT

Grown-in defects such as voids and oxide precipitates critically affect the performance of Czochralski (CZ) silicon wafers. In this work, the effects of tin (Sn)-doping on vacancy-related defect evolution in CZ silicon are investigated using first-principles calculations, molecular dynamics simulations, rapid thermal annealing, gold diffusion, and deep-level transient spectroscopy. The results show that Sn-doping fundamentally alters the kinetic evolution pathways of vacancy-related defects during crystal cooling under low intrinsic vacancy concentrations relevant to practical CZ growth. Substantial SnV complex formation initiates within the temperature range of 1100–1080 °C and continues down to ~1030 °C. Since this temperature window substantially overlaps with the regime for vacancy aggregation and void evolution, the continuous trapping of mobile vacancies by substitutional Sn atoms kinetically competes with void nucleation and growth during cooling. As the temperature further decreases below ~1030 °C, part of the SnV complexes react with interstitial oxygen atoms to form relatively stable SnVO complexes. Importantly, although SnVO complexes contain vacancy constituents, they exhibit negligible binding affinity for additional oxygen atoms and, therefore, cannot act as effective heterogeneous nucleation centers for oxygen precipitation. Meanwhile, the reduced population of mobile free vacancies further suppresses the formation of conventional VO₂ complexes acting as oxygen precipitation nuclei. Consequently, the formation of both voids and grown-in oxide precipitates is significantly suppressed in Sn-doped CZ silicon. This work reveals a thermodynamic-kinetic decoupling in vacancy behavior induced by Sn-doping: although Sn-doping lowers the vacancy formation energy, it simultaneously redirects mobile vacancies into kinetically inactive configurations through competing kinetic pathways. More broadly, this work establishes a kinetic framework for understanding impurity-mediated vacancy evolution and defect engineering in CZ silicon under realistic crystal growth conditions.

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I. INTRODUCTION

Czochralski (CZ) silicon is a base material for the microelectronics industry, where device yield is critically dependent on crystal quality. The cooling process during CZ silicon crystal growth typically leads to the formation of grown-in defects, primarily voids and oxide precipitates, which are key determinants of final crystal quality.^{1–3} Consequently, the engineering of these grown-in defects has long been a central focus in CZ silicon research and development. Under vacancy-rich growth conditions,

voids form via the aggregation of excess vacancies, while vacancy (V)-interstitial oxygen (O_i) interactions yield V-O (e.g., VO₂) complexes that serve as precursors for oxide precipitate nucleation.³ One established approach to defect engineering involves controlling key crystal growth parameters, such as the temperature gradient at the crystal/melt interface and the pulling rate.^{1,2} Alternatively, the grown-in defects can be engineered by intentional doping with the impurities that interact with vacancies and O_i atoms within specific temperature ranges.^{4–8} This is exemplified

by nitrogen doping, which enhances oxygen precipitation and reduces void size.^{4–6} In principle, the interaction between vacancies and impurity atoms is particularly strong when the impurity has a significantly larger covalent radius than silicon.^{9,10} Given this strong interaction propensity, doping with isovalent tin (Sn), with a substantially larger covalent radius, represents a promising strategy for engineering the grown-in defects, which motivates the present in-depth study.

The covalent radius of Sn atoms (1.40 Å) is larger than that of Si atoms (1.17 Å). Therefore, Sn atoms can act as efficient vacancy trapping sites in silicon. The formation of SnV complexes through Sn–V interactions was reported as early as the 1970s. Brelot first observed via room-temperature infrared (IR) absorption spectroscopy that in the electron-irradiated CZ silicon with Sn concentration $[\text{Sn}]$ of $5 \times 10^{18} \text{ cm}^{-3}$, the Sn atoms directly trap the irradiation-induced vacancies to form SnV complexes, reducing the formation rate of oxygen-vacancy pairs (A centers) by a factor of 13.¹¹ Watkins demonstrated through low-temperature irradiation and annealing that the isolated vacancies migrated and were trapped by the Sn atoms upon the annealing to 180 K, forming SnV complexes.¹² Using low-dose (1×10^{14} – $3 \times 10^{15} \text{ cm}^{-2}$) 2 MeV electron irradiation on n-type CZ silicon with $[\text{Sn}]$ ranging from 3 to $6 \times 10^{18} \text{ cm}^{-3}$, Svensson *et al.* reported that the Sn atoms captured the vacancies five times more efficiently than the O_i atoms, further confirming the effective formation of SnV at room temperature.¹³ Nylandsted Larsen *et al.* investigated 2 MeV electron-irradiated n-type float zone (FZ) silicon with Sn-doping, suggesting that a $[\text{Sn}]$ of 10^{17} – 10^{18} cm^{-3} was required to effectively form SnV complexes.¹⁴ Chroneos *et al.* experimentally observed SnVO complexes in CZ silicon with high $[\text{Sn}]$ ($9 \times 10^{18} \text{ cm}^{-3}$) using Fourier transform infrared (FTIR) spectroscopy, identifying a characteristic IR absorption peak at 806 cm^{-1} and a formation temperature of $\sim 400^\circ\text{C}$. Moreover, their density functional theory (DFT) calculations revealed the stable configuration of SnVO, showing that the binding energy was higher for Sn near the vacancy (SnVO) than near the oxygen (SnOV) because the attraction between Sn and the vacancy slightly exceeded the repulsion between Sn and O_i atoms. Furthermore, they found that the formation of SnVO complexes significantly inhibited the transformation from VO to VO_2 .¹⁵ Earlier work by Khirunenko *et al.* had also implied the existence of SnVO complexes.¹⁶ Under high-dose irradiation, the vacancies released from the dissociation of SnV complexes combined with other SnV or divacancies (V_2) to form SnV_2 complexes, as observed by Fanciulli and Byberg via electron paramagnetic resonance (EPR) spectroscopy in the electron-irradiated samples with $1 \times 10^{18} \text{ e}^-/\text{cm}^2$.¹⁷

Sn-doping also significantly influences the aggregation and precipitation behavior of oxygen in CZ silicon. Neimash *et al.* reported that in n-type CZ silicon, Sn-doping (1.74×10^{18} – $6.51 \times 10^{18} \text{ cm}^{-3}$) reduced the formation rate of oxygen thermal donors (OTDs).¹⁸ Similarly, Angeletos *et al.* found that Sn-doping decreased the reaction rate and increased the activation energy for oxygen aggregation during isothermal annealing at 450°C .¹⁹ In contrast, the studies on p-type CZ silicon by Wijaranakula showed that Sn-doping ($2.5 \times 10^{18} \text{ cm}^{-3}$) enhanced oxygen precipitation within the temperature range of 700 – 750°C .^{20,21} The interaction of Sn with other impurities further complicated its effects. For

instance, Sgourou *et al.* observed that in the electron-irradiated CZ silicon, carbon doping alone suppressed the formation of spherical oxide precipitates and promoted platelet ones, whereas co-doping with Sn reversed this trend, leading to the dominance of spherical precipitates.²² Gao *et al.* found that the effect of Sn-doping on oxygen precipitation in CZ silicon depended on the vacancy concentration. Namely, Sn-doping suppressed oxygen precipitation under vacancy-deficient conditions, while it promoted oxygen precipitation under vacancy-rich conditions.²³ Subsequent studies have further advanced the understanding of Sn-related defect engineering in silicon. Chroneos *et al.* employed mass-action analysis to calculate the relative concentrations of VO defects under different isovalent doping concentrations from a thermodynamic equilibrium perspective, concluding that Sn or Pb doping at concentrations above 10^{18} cm^{-3} is an effective strategy for suppressing VO defects.²⁴ Moreover, in a recent review, they systematically summarized the influence of isovalent doping, including Sn-doping, on the formation and evolution of oxygen- and carbon-related defects in electron-irradiated CZ silicon, further highlighting the important role of Sn–vacancy interactions in modulating irradiation-induced defects.²⁵ Abdurrazzaq *et al.* investigated the effect of isovalent doping on the electrical activity of hydrogen-passivated VO-related defect complexes.²⁶ More recently, Kuganathan *et al.* performed spin-polarized DFT calculations to systematically examine the structures and binding energies of SnV and SnV_2 clusters, further confirming the strong binding of Sn with vacancies in SnV- and SnV_2 -type configurations.²⁷

Despite the aforementioned advances, most direct experimental knowledge of Sn-related vacancy complexes in silicon has been derived from electron-irradiated materials with artificially elevated vacancy concentrations. Under such conditions, SnV and SnVO complexes can reach concentrations detectable by spectroscopic techniques, and their formation and dissociation behaviors have been characterized accordingly.¹⁵ However, in practical CZ crystal growth, the intrinsic vacancy concentration is several orders of magnitude lower, and the system evolves under continuous high-temperature cooling rather than isothermal conditions. Whether the defect reactions established in irradiation studies remain applicable under these fundamentally different conditions remains an open question. Although the previous work by Gao *et al.* established the vacancy-concentration-dependent effect of Sn-doping on oxygen precipitation,²³ the temperature-dependent SnV/SnVO formation pathways under low-vacancy, continuous-cooling conditions relevant to practical CZ growth remain unresolved. This issue is particularly important for understanding how Sn-doping can affect the formation of both voids and grown-in oxide precipitates. More importantly, the formation of vacancy-related complexes during crystal cooling is inherently a kinetic problem involving the competition between multiple reaction pathways. In particular, vacancy capture by substitutional Sn atoms and V–O pairing represent two distinct mechanisms with different kinetic requirements. The former proceeds via a trap-limited process involving mobile vacancies and immobile Sn atoms, whereas the latter requires the encounter of two mobile species. This distinction suggests that the evolution of vacancy populations in Sn-doped CZ silicon cannot be understood solely in terms of defect energetics, but must be analyzed within a kinetic framework.

To address these issues, this work moves beyond the high-vacancy, low-temperature regime typical of irradiation studies to reveal the defect engineering mechanism of Sn-doping under the low intrinsic vacancy concentrations and high-temperature cooling conditions characteristic of industrial crystal growth. It systematically investigates the formation and evolution of SnV and SnVO complexes under conditions directly relevant to CZ crystal growth, by combining first-principles calculations, molecular dynamics (MD) simulations, and carefully designed experiments. By correlating theoretical predictions with experimental observations, we establish temperature-resolved pathways for vacancy evolution and elucidate how Sn-doping redistributes vacancy-related species during the cooling process. On this basis, a physically consistent mechanism is proposed to explain the simultaneous suppression of void and grown-in oxide precipitate formation in Sn-doped CZ (SCZ) silicon.

II. EXPERIMENTAL

Two lightly phosphorus-doped, n-type $\langle 100 \rangle$ -oriented silicon crystal ingots, each 150 mm in diameter, were grown by the CZ method under nearly identical conditions. One ingot was intentionally doped with Sn and is hereafter referred to as SCZ silicon; the other, undoped ingot serves as the conventional CZ reference. To ensure comparable thermal history, resistivity, and interstitial oxygen concentration $[(O_i)]$, three wafers were consecutively sliced from corresponding axial positions of each ingot. All wafers were double-side polished to a final thickness of approximately 1.3 mm. After the oxygen thermal donors were eliminated by 650 °C/0.5 h annealing, the resistivity of all wafers was $\sim 20 \Omega \text{ cm}$, measured by the four-point probe method. Additionally, the $[O_i]$ in all wafers was $\sim 7.0 \times 10^{17} \text{ cm}^{-3}$, measured by Fourier transform infrared spectroscopy (FTIR, Bruker IFS 66 V) with a conversion factor of $3.14 \times 10^{17} \text{ cm}^{-2}$ (ASTM F121-89). The substitutional carbon concentration in all wafers was below the detection limit of an FTIR spectrometer. The [Sn] in the SCZ silicon wafers was $\sim 7.0 \times 10^{17} \text{ cm}^{-3}$, measured by secondary ion mass spectrometry (SIMS) using a CAMECA IMS 4f system. Multiple samples with dimensions of $1.5 \times 1.5 \text{ cm}^2$ were cleaved from the CZ and SCZ silicon wafers for the annealing schemes serving different purposes. Here, the sampling locations excluded the regions within 3 cm from the wafer edge. Furthermore, the samples sized $2.0 \times 4.0 \text{ cm}^2$ were, respectively, cleaved from the central region of one CZ and one SCZ silicon wafer for delineating the flow pattern defects (FPDs), which correspond to the voids, by Secco preferential etching.

To investigate the interaction between Sn atoms and vacancies, rapid thermal annealing (RTA), gold (Au) diffusion, and deep-level transient spectroscopy (DLTS) characterization were employed. Twelve CZ and twelve SCZ silicon samples were first subjected to pre-anneal at 1250 °C for 2 h under an argon (Ar) atmosphere to substantially eliminate the grown-in oxide precipitates. In the subsequent RTA under an Ar atmosphere, each sample was heated from room temperature to 1250 °C at a rate of 100 °C/s and then held for 60 s. For the subsequent cooling process, the sample was cooled at a rate of 5 °C/s to a temperature of 1100, 1080, 1050, 1020, 950, or 800 °C, after which the sample

was immediately withdrawn from the RTA furnace for air-cooling. The measurements showed that the cooling rate from 1100, 1080, 1050, 1020, or 950 °C to 800 °C in air was approximately 100 °C/s. All samples after the above RTA were immersed in a diluted HF solution [HF (49%): $\text{H}_2\text{O} = 1:10$, volume ratio] for 5 min to remove any possible oxide layer. Subsequently, all samples were equally divided into two groups, each containing six CZ and six SCZ silicon samples. One group was subjected to mechanochemical polishing, removing a thickness of $\sim 300 \mu\text{m}$ from each side of a sample to eliminate the vacancy out-diffusion zones generated by the RTA. The other group was first subjected to Au diffusion as follows: (1) a $\sim 120 \text{ nm}$ thick Au film was sputtered onto one side of each sample; (2) the samples were annealed in a quartz tube furnace under an Ar atmosphere at 750 °C for 5 h to allow Au atoms to diffuse into the sample bulk. Likewise, this group of samples after Au diffusion was also subjected to mechanochemical polishing, removing a thickness of $\sim 300 \mu\text{m}$ from each side of a sample to remove the residual Au film on the surface and the vacancy out-diffusion zones generated by the RTA. Both groups of samples above were subjected to DLTS characterization. For this purpose, a $\sim 150 \text{ nm}$ thick Au film was sputtered onto one side of each sample to form a Schottky contact, and an indium-gallium (In-Ga) alloy layer was applied to the other side to form an ohmic contact. DLTS characterization was performed on a PhysTech Fourier Transform DLTS system, with the specific measurement parameters as follows: temperature range 100–300 K, reverse bias $V_R = -6 \text{ V}$, fill pulse voltage $V_P = -0.1 \text{ V}$, sampling period $t_w = 204.8 \text{ ms}$ (during the bias pulse from V_R to V_P), and the fill pulse duration $t_p = 10 \text{ ms}$.

To examine the effect of Sn-doping on oxygen precipitation, the comparative CZ and SCZ silicon samples were subjected to identical annealing treatments under an Ar atmosphere. The following annealing schemes were employed: (1) 1250 °C/2 h + 800 °C/4 h + 1000 °C/0–16 h, (2) ramping from 800 to 1000 °C at a rate of 1 °C/min and holding for up to 80 h, (3) isothermal annealing at 1000 or 1050 °C for up to 80 h, or at 1100 °C for up to 48 h. After each high-temperature anneal at 1000, 1050, or 1100 °C, the $[O_i]$ in each sample was measured by FTIR spectroscopy. Prior to the measurement, any surface oxide was removed by etching in the diluted HF solution described earlier. Furthermore, to characterize the microstructure and size of the formed oxide precipitates, the representative SCZ and CZ silicon samples subjected to 1000 °C/80 h annealing were thinned to electron transparency by successive mechanical thinning and ion-milling, followed by transmission electron microscopy (TEM) characterization with an FEI Tecnai G2 F20 S-Twin transmission electron microscope operated at 200 kV.

The densities of grown-in oxygen precipitates in SCZ and CZ silicon wafers were also investigated indirectly by examining copper precipitation behavior. Two sets of SCZ and CZ silicon samples were prepared. One set was pre-annealed at 1250 °C for 2 h to substantially dissolve the grown-in oxide precipitates, while the other set was kept as-received to retain the grown-in oxide precipitates. All samples were then immersed in a 1 mol/l CuCl_2 solution for 10 min to introduce copper, rinsed with de-ionized water, and annealed at 1000 °C for 5 min under an Ar atmosphere, followed by cooling in the flowing air to induce copper precipitation.

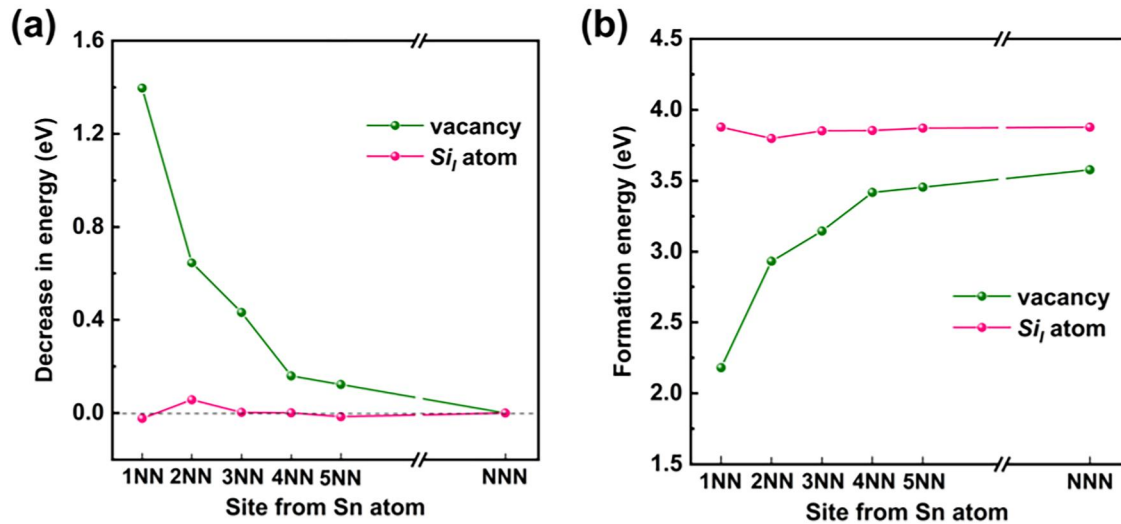


FIG. 1. (a) Decreases in energy for a vacancy or an interstitial silicon atom moving from a distant (NNN) site to the 1st to 5th nearest neighboring sites of a substitutional Sn atom along the $\langle 110 \rangle$ direction, (b) formation energies of a vacancy or an interstitial silicon atom at the 1st to 5th nearest neighboring sites of a substitutional Sn atom. Here, NNN denotes the reference configuration where the vacancy is infinitely far from the Sn atom, corresponding to a pristine silicon supercell.

To reveal the copper precipitates, the samples were etched for 5 min using Yang1 etchant [HF (49%): CrO_3 (0.5M) = 1:1, volume ratio] and observed with an Olympus MX-50 optical microscope. For microstructural analysis, the selected CZ silicon samples from both groups were thinned to electron transparency by successive mechanical polishing and ion-milling and examined using a FEI Tecnai G2 F20 S-Twin transmission electron microscope operated at 200 kV.

The void formation in CZ and SCZ silicon wafers was assessed by characterizing the FPDs, a well-established etching method for revealing the vacancy-type defects. The as-received CZ and SCZ silicon samples ($2.0 \times 4.0 \text{ cm}^2$) were vertically immersed in Secco etchant [$\text{K}_2\text{Cr}_2\text{O}_7$ (0.15M):HF (49%) = 1:2, volume ratio] for 30 min without stirring. After the etching, the samples were rinsed with de-ionized water, dried under nitrogen flow, and examined using an Olympus MX-50 optical microscope at $50\times$ magnification. The FPD densities were determined by counting the FPDs in 20 uniformly spaced regions on each sample surface.

III. THEORETICAL CALCULATIONS

To understand the interactions among the vacancies and various impurity atoms in silicon, we performed a series of first-principles calculations. Both DFT calculations and MD simulations were performed using the VASP software package,^{28,29} adopting the generalized gradient approximation (GGA) in the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional.^{30,31} For DFT calculations, a supercell containing 216 silicon atoms named Si_{216} was first established. On the basis of this supercell, the supercells including Si_{215}V , Si_{216}I , $\text{Si}_{215}\text{Sn}_s$, $\text{Si}_{216}\text{O}_i$, $\text{Si}_{214}\text{Sn}_s\text{VO}$, $\text{Si}_{214}\text{Sn}_s\text{VO}_2$, Si_{215}VO , $\text{Si}_{214}\text{Sn}_s(\text{n})\text{V}$, $\text{Si}_{215}\text{Sn}_s(\text{n})\text{I}$, $\text{Si}_{214}\text{SnVAu}_i$, and $\text{Si}_{214}\text{SnVOAu}_i$ were constructed, respectively.

Among them, V, I, O_i , Sn_s , and Au_i represent vacancy, self-interstitial silicon atom, interstitial oxygen atom, substitutional Sn atom, and interstitial Au atom, respectively. $\text{Si}_{214}\text{Sn}_s(\text{n})\text{V}$ represents a series of supercells where the vacancy is located at the n (1st–5th) nearest neighboring site of the substitutional Sn atom along the $\langle 110 \rangle$ direction. The above supercells were structurally optimized to obtain the corresponding supercell energies. In static DFT calculations, a $2 \times 2 \times 2$ k-point grid was used to sample the Brillouin zone, the cutoff energy after the plane wave expansion was 450 eV, the energy convergence accuracy during the structural optimization was 1.0×10^{-5} eV, and the optimization was finally performed until the force on each atom was lower than 0.01 eV/Å. For MD simulations, the NVT ensemble was used for calculations, the supercell temperature was set according to the needs of the studies, the simulation time step was set to 1 fs, the number of simulation steps was 2000, and the energy convergence accuracy during the simulation process was 5.0×10^{-5} eV. The MD simulations in this work are not intended to quantitatively reproduce long-time diffusion kinetics, but rather to reveal the relative local migration tendencies and trapping behaviors of vacancy-related defects within different atomic environments.

IV. RESULTS AND DISCUSSION

A. On theoretical calculations

1. Effect of Sn-doping on vacancy formation

Figure 1(a) presents the decreases in energy for a vacancy or an interstitial silicon (Si_i) atom moving from a distant (NNN) site to the 1st to 5th nearest neighboring sites of a substitutional Sn atom along the $\langle 110 \rangle$ direction, which are calculated using the

following expressions:

$$\Delta E_n^V = E_{\text{Si}_{215}V} + E_{\text{Si}_{215}\text{Sn}_n} - E_{\text{Si}_{214}\text{Sn}_n(n)V} - E_{\text{Si}_{216}}, \quad (1)$$

$$\Delta E_n^I = E_{\text{Si}_{216}I} + E_{\text{Si}_{215}\text{Sn}_n} - E_{\text{Si}_{215}\text{Sn}_n(n)I} - E_{\text{Si}_{216}}, \quad (2)$$

where (n) represents the 1st–5th nearest neighboring site of the substitutional Sn atom along the $\langle 110 \rangle$ direction. As seen, the energy decreases continuously as the vacancy approaches the first nearest-neighbor site of the Sn atom, indicating a strong binding interaction that leads to the formation of a SnV complex. DFT calculations yield a formation energy of 1.40 eV for this complex, confirming its high thermal stability. In contrast, the system energy remains nearly unchanged as a Si_i atom approaches the Sn atom.

Figure 1(b) shows the formation energies of vacancies and Si_i atoms at the 1st to 5th nearest neighboring sites of the Sn atom, calculated using the following expressions:

$$E_f^{V,(n)} = E_{\text{Si}_{214}\text{Sn}_n(n)V} - E_{\text{Si}_{215}\text{Sn}_n} + \frac{1}{216} E_{\text{Si}_{216}}, \quad (3)$$

$$E_f^{I,(n)} = E_{\text{Si}_{215}\text{Sn}_n(n)I} - E_{\text{Si}_{215}\text{Sn}_n} - \frac{1}{216} E_{\text{Si}_{216}}. \quad (4)$$

As shown in Fig. 1(b), the formation energy of a vacancy decreases as its position approaches the Sn atom, whereas the formation energy of a Si_i atom remains almost unchanged. This indicates that vacancy formation is facilitated near the Sn atom, while Si_i formation is hardly affected. From Fig. 1, it can be inferred that Sn-doping influences the vacancy behavior in silicon in two ways. First, it reduces the vacancy formation energy, which would thermodynamically increase the equilibrium vacancy concentration. Second, and more importantly, the energy minimum at the first nearest neighboring site drives the preferential binding of vacancies to Sn atoms below a certain temperature, leading to the formation of stable SnV complexes, as will be confirmed experimentally. Therefore, even with a lower vacancy formation energy, Sn-doping results in a lower concentration of free vacancies because the vacancies are effectively trapped in SnV complexes.

2. Formation of SnV complexes

Figure 2(a) shows a magnified local view of the optimized configuration at 0 K of a silicon supercell containing one substitutional Sn atom and one vacancy, with the vacancy initially located at the second-nearest-neighbor site of the Sn atom. Figures 2(b)–2(d) show representative local configurations after a 2 ps MD simulation at 800, 1070, and 1080 °C, respectively. At all the simulated temperatures up to 1070 °C, the vacancy migrates to the first-nearest-neighbor site of the Sn atom, forming a stable SnV complex; representative configurations at 800 and 1070 °C are shown in Figs. 2(b) and 2(c). In contrast, at 1080 °C [Fig. 2(d)], the vacancy moves away from the Sn atom, indicating that SnV complex formation is inhibited. The MD simulations, therefore, place the upper-temperature boundary for stable SnV complex formation between 1070 and 1080 °C. Because this range overlaps

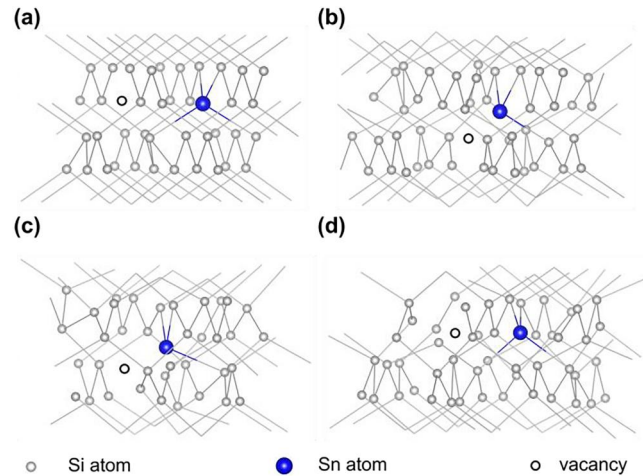


FIG. 2. Magnified local views of (a) the optimized configuration at 0 K of a silicon supercell containing a vacancy initially at the second-nearest-neighbor site of a substitutional Sn atom and (b)–(d) the configurations after a 2 ps MD simulation at 800, 1070, and 1080 °C, respectively. Only the local atomic environment around the Sn atom and vacancy is shown for clarity.

with the typical void-formation temperature in CZ silicon (1020–1100 °C),³ SnV complex formation is expected to directly influence void formation in SCZ silicon.

Figure 3(a) shows a magnified local view of the optimized configuration at 0 K of a silicon supercell containing one interstitial oxygen (O_i) atom and one vacancy, with the vacancy initially placed at the second-nearest-neighbor site of the O_i atom. Figures 3(b)–3(d) show the corresponding local configurations

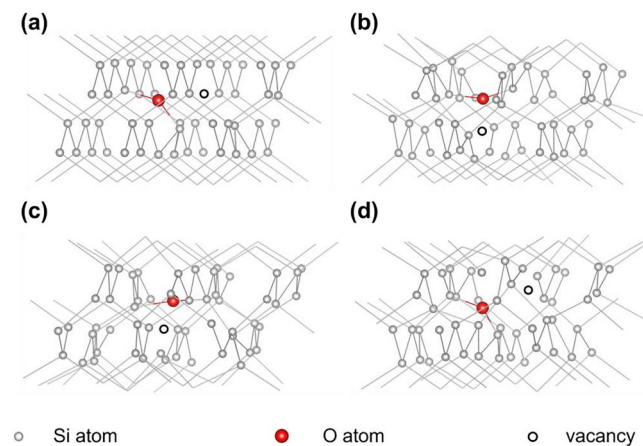


FIG. 3. Magnified local views of (a) the optimized configuration at 0 K of a silicon supercell containing a vacancy initially at the second-nearest-neighbor site of an interstitial oxygen atom and (b)–(d) the configurations after a 2 ps MD simulation at 800, 1030, and 1050 °C, respectively. Only the local atomic environment around the oxygen atom and vacancy is shown for clarity.

after a 2 ps MD simulation at 800, 1030, and 1050 °C, respectively. At 800 and 1030 °C [Figs. 3(b) and 3(c)], the vacancy and the O_i atom migrate toward each other and combine to form a VO complex. In contrast, at 1050 °C [Fig. 3(d)] and at the higher simulated temperatures, no stable VO complex is observed, indicating that VO formation occurs below approximately 1050 °C. The DFT calculations give binding energies of 1.69 eV for the VO complex and 1.40 eV for the SnV complex, indicating that VO complex formation is thermodynamically more favorable. From a kinetic perspective, however, VO formation requires the encounter and association of two highly mobile species, whereas substitutional Sn atoms have a high diffusion barrier (~ 3.5 eV) and are essentially immobile.³² Consequently, SnV complex formation is governed primarily by vacancy migration toward nearly immobile substitutional Sn atoms, a process that is kinetically more favorable than VO complex formation, which requires the migration, encounter, and association of a vacancy and an interstitial oxygen atom. This is consistent with the MD results showing that the SnV complexes form up to ~ 1070 °C (Fig. 2), whereas the VO complexes form only below ~ 1050 °C (Fig. 3). Thus, during the cooling process of SCZ silicon crystal growth, substantial SnV complex formation occurs within a higher temperature regime than VO complex formation, thereby influencing the subsequent formation of grown-in oxide precipitates.

3. Formation of SnVO complexes

Figure 4(a) shows a magnified local view of the optimized configuration at 0 K of a silicon supercell containing an SnVO complex, with an O_i atom at the first-nearest-neighbor site of an SnV complex. The calculated binding energy between the SnV complex and the O_i atom is 1.14 eV, confirming the stability of the SnVO complex, in agreement with the experimental results of Chronos *et al.*³³ Figure 4(b) shows the optimized initial

configuration at 0 K with the O_i atom located at the second-nearest-neighbor site of the vacancy in an SnV complex. Figures 4(c) and 4(d) show representative local configurations after a 2 ps MD simulation at 1020 and 1030 °C, respectively. At 1020 °C [Fig. 4(c)], the O_i atom migrates to the vacancy site and forms a stable SnVO complex, whereas no such migration occurs at 1030 °C [Fig. 4(d)]. Together with the simulations at 1000 and 1050 °C, these results indicate that SnVO complex formation is limited to temperatures below approximately 1030 °C. The SnVO and VO complexes, therefore, have similar formation temperatures. A critical distinction, however, lies in their ability to bind additional O_i atoms. The VO complex can incorporate a second O_i atom to form a VO_2 complex, which can act as an effective heterogeneous nucleation site for oxygen precipitation.³⁴ In contrast, the binding energy for adding an O_i atom to the SnVO complex is only 0.01 eV, meaning that the SnVO complex is essentially inert toward further oxygen binding and cannot serve as a heterogeneous nucleation site. As established earlier, the SnV complexes form earlier and capture the free vacancies, thereby suppressing the VO formation. As a result, the VO_2 complexes, which evolve from the VO complexes, are far outnumbered by the SnVO complexes in SCZ silicon.

4. Interaction of SnV and SnVO complexes with interstitial Au atoms

In this work, to indirectly probe the residual vacancy concentrations in the RTA-treated CZ and SCZ silicon wafers, we employed a method combining gold diffusion with DLTS. Consistent with the previous report,³⁴ we assume that the residual vacancies in the RTA-treated CZ wafers are predominantly in the form of VO_2 complexes. We note, however, that Akhmetov *et al.* detected a considerable concentration ($\sim 1.4 \times 10^{13} \text{ cm}^{-3}$) of VO_4 complexes in CZ silicon under the RTA conditions (1250 °C/30 s, Ar/ O_2 atmosphere) using the sensitivity-enhanced FTIR.³⁵ Our adoption of the VO_2 model here is a simplifying assumption for this study and does not contradict the above finding. As established previously,³⁶ the interstitial gold (Au_i) atoms can react with the VO_2 complexes at certain temperatures to form substitutional gold (Au_s) atoms. In the RTA-treated SCZ silicon wafers, however, the residual vacancies may exist as the SnV or SnVO complexes. A key question then arises: can the in-diffused Au_i atoms similarly react with SnV or SnVO complexes to form Au_s atoms? We address this question first through MD simulations.

Figures 5(a) and 5(b) show magnified local views of the optimized configurations at 0 K of silicon supercells containing an Au_i atom initially located at the second-nearest-neighbor site of the vacancy in an SnV and an SnVO complex, respectively. Figures 5(c) and 5(d) show the corresponding local configurations after a 2 ps MD simulation at 750 °C. At this temperature, both the SnV and SnVO complexes remain stable. In the SnV-containing supercell [Fig. 5(c)], the Au_i atom migrates to the vacancy and becomes a substitutional Au (Au_s) atom. In contrast, in the SnVO-containing supercell [Fig. 5(d)], the Au atom remains interstitial. This difference arises because the combined presence of Sn and O_i in the SnVO complex reduces the effective space around the vacancy relative to that in the SnV complex, thereby hindering

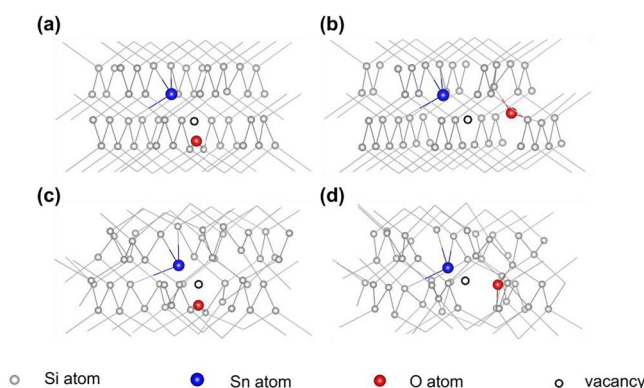


FIG. 4. Magnified local views of (a) the optimized configuration at 0 K of a silicon supercell containing a stable SnVO complex; (b) the optimized initial configuration at 0 K with an interstitial oxygen atom at the second-nearest-neighbor site of the vacancy in an SnV complex; and the configurations after a 2 ps MD simulation at (c) 1020 and (d) 1030 °C. Only the local atomic environment around the Sn atom, oxygen atom, and vacancy is shown for clarity.

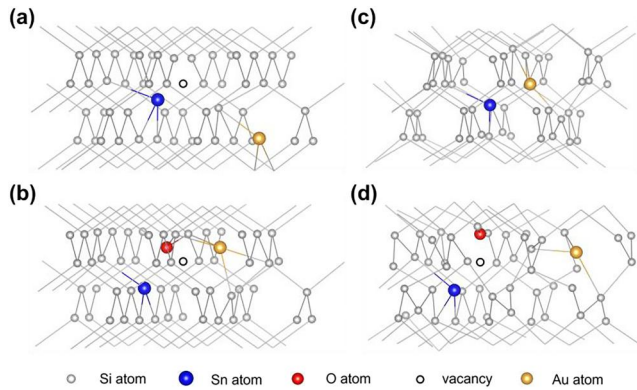


FIG. 5. Magnified local views of the optimized configurations at 0 K of silicon supercells containing an interstitial Au atom initially at the second-nearest-neighbor site of the vacancy in (a) an SnV complex and (b) an SnVO complex, and the corresponding configurations after a 2 ps MD simulation at 750 °C for (c) the SnV-containing system and (d) the SnVO-containing system. Only the local atomic environment around the defect complexes is shown for clarity.

Au_i incorporation. The MD simulations, therefore, demonstrate that Au_i atoms can react with SnV complexes to form Au_s atoms but do not undergo the same reaction with SnVO complexes.

B. On experimental studies

1. Formation of SnV and SnVO complexes

Figure 6(a) shows the DLTS spectra of the CZ and SCZ silicon wafers subjected to the RTA soaking stage at 1250 °C for

60 s with the subsequent cooling at a rate of 5 °C/s to 1080 °C, immediately followed by air-cooling to room temperature. As seen, the DLTS spectrum of the CZ silicon wafer exhibits no significant characteristic signal peaks, whereas that of the SCZ silicon wafer shows a distinct signal peak near 160 K. Clearly, this signal peak corresponds to a Sn-related defect. The energy level corresponding to the aforementioned signal peak is fitted to be $E_c - 0.28$ eV, which is close to the energy level ($E_c - 0.29$ eV) of the SnV complex, as reported by Simoen *et al.* in the electron-irradiated tin-doped silicon wafers.³⁷ Therefore, it is believed that the DLTS peak near 160 K for the SCZ silicon wafer shown in Fig. 6(a) is associated with the SnV complex. Figure 6(b) presents the DLTS spectra of the SCZ silicon wafers subjected to the RTA soaking stage at 1250 °C for 60 s with the subsequent cooling at a rate of 5 °C/s to different terminal temperatures of 1100, 1080, 1050, 1020, 950, and 800 °C, respectively, immediately followed by air-cooling to room temperature. It can be seen that only when the RTA terminal temperature is 1080, 1050, or 1020 °C does the DLTS spectrum exhibit significant signal peaks associated with the SnV complex. By comparison, the DLTS peak of SnV complex is the strongest in the case of a terminal temperature of 1080 °C. Notably, when the RTA terminal temperature is 1100 °C, the DLTS peak of the SnV complex is considerably weak. The above results indicate that the formation of SnV complexes mainly occurs during the cooling process of RTA, rather than during the high-temperature soaking stage. Furthermore, it can be inferred that the onset formation of SnV complexes occurs at a certain temperature between 1100 and 1080 °C. It should be noted that this experimentally inferred onset range does not strictly correspond to the onset temperature (1080–1070 °C) obtained from the MD simulations. In the MD simulations, the onset temperature reflects the upper limit at which a vacancy can still migrate toward and become trapped by a substitutional Sn atom within

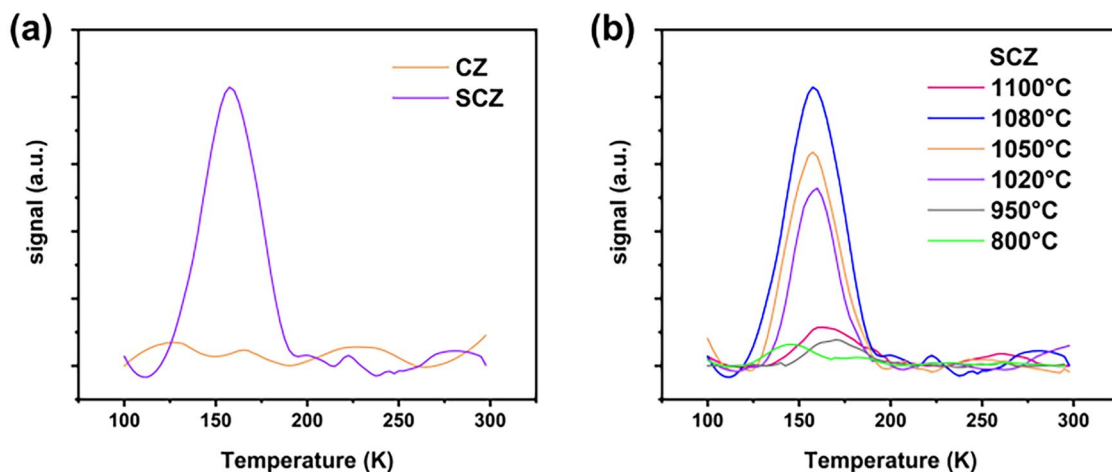


FIG. 6. (a) DLTS spectra of CZ and SCZ silicon samples subjected to the RTA soaking stage at 1250 °C for 60 s, followed by cooling at a rate of 5 °C/s to 1080 °C, immediately followed by the air-cooling out of the RTA furnace to room temperature, (b) DLTS spectra of SCZ silicon samples subjected to the same RTA soaking stage, followed by cooling at a rate of 5 °C/s to 1100, 1080, 1050, 1020, 950, and 800 °C, respectively, followed by withdrawal from the RTA furnace and air-cooling to room temperature.

the picosecond simulation timescale of an idealized local configuration. In contrast, the DLTS results reflect the accumulation and freeze-in of a detectable population of SnV complexes during continuous cooling under non-equilibrium conditions. Therefore, the experimentally inferred temperature range is expected to be somewhat broader than the atomistic transition range predicted by MD simulations. Nevertheless, both results consistently support that substantial SnV complex formation occurs within the high-temperature regime relevant to void evolution.

It is also seen from Fig. 6(b) that when the RTA terminal temperature is 950 and 800 °C, the signal peaks become nearly negligible in the DLTS spectra. Nevertheless, it cannot be concluded that the SnV complexes do not form under these conditions. This behavior may originate from two possibilities. One is that the concentration of SnV complexes retained during the RTA cooling process falls below the detection limit of DLTS. The other is that part of the SnV complexes continue to evolve into other vacancy-related complexes during the subsequent cooling process, thereby becoming undetectable by direct DLTS measurements. In fact, the aforementioned MD studies have revealed that the SnV complexes can further interact with the O_i atoms to form SnVO complexes within the temperature range of approximately 1050–1020 °C. Therefore, the DLTS measurements in Fig. 6 alone are insufficient to distinguish whether the weakening or disappearance of the SnV-related signal at lower terminal temperatures originates primarily from insufficient SnV retention or from the subsequent transformation of SnV into other complexes such as SnVO. To further resolve the evolution pathways of vacancy-related defects during cooling, an additional experimental probe capable of tracking the retention and redistribution of vacancy-related species is required. For this purpose, Au diffusion combined with DLTS characterization was further employed in the following study,

because the conversion of Au_i atoms into Au_s atoms is highly sensitive to the availability and chemical state of vacancy-related defects. By correlating the resulting Au_s concentration $[Au_s]$ evolution with the MD predictions, the sequential transformation from free vacancies to SnV and subsequently to SnVO complexes can be further clarified.

Figure 7(a) shows the DLTS spectra of the CZ and SCZ silicon samples subjected to the RTA soaking stage at 1250 °C for 60 s with subsequent cooling at a rate of 5 °C/s to 1080 °C, immediately followed by the air-cooling to room temperature, and then subjected to Au diffusion at 750 °C for 5 h. Unlike Fig. 6, which directly probes the presence of detectable SnV complexes, the Au-assisted DLTS measurements in Fig. 7 are intended to track the evolution and redistribution of the overall vacancy-related population during cooling. It can be seen that the DLTS spectra of both silicon wafers exhibit a distinct signal peak near 245 K. The energy level corresponding to this signal peak is fitted to be $E_c - 0.55$ eV, which is close to the energy level ($E_c - 0.56$ eV) of Au_s atoms in silicon.^{38,39} Therefore, the DLTS peak near 245 K in Fig. 7(a) can be considered to be related to Au_s atoms. During the Au diffusion process, the Au impurities enter the silicon wafer bulk in the form of Au_i atoms, and a portion of these Au_i atoms can interact with the vacancies (including free vacancies and vacancy-related complexes) to form Au_s atoms. Via the free vacancies, VO_2 complexes in CZ silicon, and SnV complexes in SCZ silicon as the mediators, the reaction equations for the conversion of Au_i atoms into Au_s atoms can be expressed as follows:

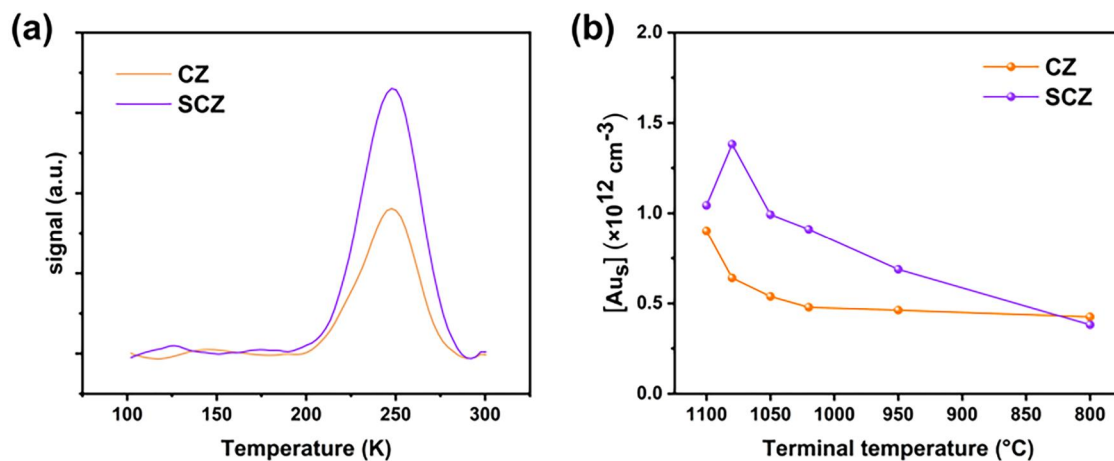
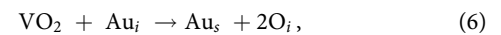
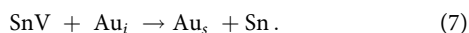


FIG. 7. (a) DLTS spectra of CZ and SCZ silicon samples subjected to the RTA soaking stage at 1250 °C for 60 s, followed by cooling at a rate of 5 °C/s to 1080 °C, immediately followed by the air-cooling out of the RTA furnace to room temperature, and subsequently subjected to Au diffusion at 750 °C for 5 h; (b) concentrations of Au_s atoms measured by DLTS in CZ and SCZ silicon samples subjected to the same RTA soaking stage, followed by cooling at a rate of 5 °C/s to 1100, 1080, 1050, 1020, 950, and 800 °C, respectively, immediately followed by withdrawal from the RTA furnace and air-cooling to room temperature, and subsequently subjected to Au diffusion at 750 °C for 5 h.



From the above three reaction equations, it is known that $[\text{Au}_s]$ is positively correlated with the concentration of V, VO_2 , or SnV complexes. Therefore, we can indirectly compare the concentration of the retained vacancies (including the free vacancies and vacancy-related complexes) during the RTA cooling process by comparing $[\text{Au}_s]$ in different silicon samples. It should be noted that the use of $[\text{Au}_s]$ as a proxy for the retained vacancy population involves several simplifying assumptions. First, the conversion of Au_i to Au_s is assumed to proceed efficiently via the reactions with vacancy-related defects, including free vacancies, VO-related complexes, and SnV complexes. Second, the vacancy-related complexes that are sterically or energetically unfavorable for Au incorporation, such as SnVO, are not expected to contribute to the measured $[\text{Au}_s]$. Third, possible contributions from higher-order V-O complexes (e.g., VO_n with $n \geq 3$) are not explicitly distinguished in the present analysis. Despite these limitations, the comparative trends in $[\text{Au}_s]$ between CZ and SCZ silicon samples under identical thermal histories provide a robust semi-quantitative measure of the relative evolution of vacancy populations. In particular, since all samples undergo identical Au diffusion conditions, systematic differences in $[\text{Au}_s]$ can be reliably attributed to the differences in the retention and transformation of vacancy-related defects during the preceding RTA cooling process.

Figure 7(a) shows that the intensity of the DLTS peak related to Au_s atoms in the SCZ silicon is significantly greater than that in the CZ silicon, indicating that the retained vacancy-related population in the SCZ silicon is significantly higher than that in the CZ silicon. This is because during the RTA cooling process from 1250 to 1080 °C, tin atoms in the SCZ silicon combine with the free vacancies to form SnV complexes. The formation of SnV complexes converts the fast-diffusing free vacancies into the slowly diffusing complexes, effectively reducing the out-diffusion of vacancies to the surface, thus resulting in a higher retained vacancy-related population in the SCZ silicon than in the CZ silicon. Furthermore, Fig. 7(b) presents $[\text{Au}_s]$ values in the CZ and SCZ silicon samples subjected to the RTA soaking stage at 1250 °C for 60 s with subsequent cooling at a rate of 5 °C/s to different terminal temperatures of 1100, 1080, 1050, 1020, 950, and 800 °C, respectively, immediately followed by the air-cooling to room temperature, and then subjected to Au diffusion at 750 °C for 5 h. Such $[\text{Au}_s]$ values are calculated from the intensity of the signal peak near 245 K in the DLTS spectra. It can be observed that as the RTA terminal temperature decreases from 1100 to 800 °C, for CZ silicon, $[\text{Au}_s]$ decreases sequentially from $\sim 9 \times 10^{11}$ to $\sim 4 \times 10^{11} \text{ cm}^{-3}$; whereas for SCZ silicon, $[\text{Au}_s]$ first increases from $\sim 1.0 \times 10^{12} \text{ cm}^{-3}$ at the terminal temperature of 1100 °C to $\sim 1.4 \times 10^{12} \text{ cm}^{-3}$ at the terminal temperature of 1080 °C, and then gradually decreases to $\sim 4 \times 10^{11} \text{ cm}^{-3}$. The above results are to be discussed in detail below.

For CZ silicon, according to the viewpoint of Falster *et al.*³⁵ the free vacancies begin to combine with the O_i atoms to form VO_2 complexes at around 1020 °C.³⁴ For the three CZ silicon samples with the RTA terminal temperatures of 1100, 1050, and 1020 °C, respectively, during the cooling period inside the RTA furnace, the vacancies within the silicon wafer bulk still exist

mainly in the form of free vacancies. After being withdrawn from the RTA furnace, since the cooling rate is approximately 100 °C/s, the time these three wafers spent passing through the temperature range for the VO_2 complex formation is similar. In fact, regardless of the concentration of VO_2 complexes formed during the air-cooling process out of the RTA furnace, the total vacancy concentrations within the bulk of these three CZ silicon samples are nearly determined by the concentration of free vacancies retained at the time of withdrawal from the RTA furnace. Obviously, as the RTA terminal temperature decreases from 1100 to 1020 °C, the cooling time inside the RTA furnace is prolonged, and the out-diffusion of free vacancies from the silicon wafer bulk becomes more significant, thereby reducing the concentration of free vacancies in the wafer bulk. Therefore, the concentration of Au_s atoms generated during the subsequent Au diffusion process for the above three samples becomes increasingly lower as the RTA terminal temperature decreases, which accounts for the experimental results presented in Fig. 7(b). When the RTA terminal temperature is below 1020 °C, during the cooling period inside the RTA furnace at temperatures below 1020 °C, the free vacancies combine with the O_i atoms to form VO_2 complexes. For the two CZ silicon samples with the RTA terminal temperatures of 950 and 800 °C, respectively, the formation of VO_2 complexes within the bulk is quite complete, and the out-diffusion of VO_2 complexes is not significant. As a result, the concentrations of VO_2 complexes in the bulk of these two samples are comparable, and hence the concentrations of Au_s atoms generated during the subsequent Au diffusion process are also comparable, as shown in Fig. 7(b).

For SCZ silicon, the evolution process of vacancies in the bulk is significantly different from that in CZ silicon. First, the vacancy formation energy in SCZ silicon (2.18 eV, at the first-nearest-neighbor site of a tin atom) is much lower than that in CZ silicon (3.58 eV), whereas the formation energies of silicon self-interstitials in the two types of wafers are nearly identical. Therefore, the concentration of vacancies introduced during the RTA soaking stage at 1250 °C for 60 s is expected to be higher in SCZ silicon than in CZ silicon. Second, according to the above DLTS characterization results, the free vacancies begin to combine with Sn atoms to form SnV complexes at a certain temperature between 1100 and 1080 °C. In other words, the formation temperature of SnV complexes is significantly higher than that of VO_2 complexes (~ 1020 °C). Therefore, under the same RTA conditions, compared with CZ silicon samples, the free vacancies in SCZ silicon samples can be captured through SnV complex formation at higher temperatures. On the basis of the above analysis, it can be concluded that, after the same RTA treatment, the total vacancy concentration in SCZ silicon is higher than that in CZ silicon. Under this premise, it becomes straightforward to understand why $[\text{Au}_s]$ in SCZ silicon is higher than that in CZ silicon shown in Fig. 7(b), except for the case in which the RTA terminal temperature is 800 °C (this exceptional case will be discussed later). In the following, we focus on how to understand the variation of $[\text{Au}_s]$ in SCZ silicon wafers as a function of the RTA terminal temperature, as shown in Fig. 7(b).

When the RTA terminal temperature is 1100 °C, the SnV complexes have not yet formed significantly during the cooling process inside the RTA furnace, and the vacancies in the silicon

wafer remain predominantly in the form of free vacancies at the time of withdrawal from the RTA furnace. During the subsequent air-cooling, although the temperature range for SnV complex formation is experienced, due to the considerably high cooling rate, the temperature decreasing from 1100 to 1020 °C takes only about 0.8 s. Under such conditions, the SnV complexes cannot form significantly, while the free vacancies can continue to out-diffuse to the surface. When the RTA terminal temperature is decreased to 1080 °C, the SnV complexes can form during the cooling period inside the RTA furnace, effectively suppressing the out-diffusion of vacancies to the surface. Moreover, the SnV complexes can be effectively retained in the wafer bulk during the subsequent cooling process. Consequently, compared with the RTA terminal temperature of 1100 °C, the RTA terminal temperature of 1080 °C results in a higher total concentration of vacancies retained in the bulk of SCZ silicon sample. However, when the RTA terminal temperature is further decreased to 1050 °C, the retained vacancies do not increase continuously. Although the wafer experiences a longer duration for the SnV complex formation during the cooling period inside the RTA furnace, in the temperature range above 1050 °C the formation and dissociation of SnV complexes are thermally activated forward–reverse reactions, while the free vacancies still possess relatively high diffusivities. In this context, during the relatively long high-temperature cooling process in the RTA furnace, even the already formed SnV complexes may dissociate to release the free vacancies again, allowing them to continue diffusing toward the surface. Meanwhile, the continuous out-diffusion of free vacancies reduces the local free vacancy concentration, thereby driving the SnV complexes to dissociate. Therefore, the condition with the RTA terminal temperature of 1050 °C is unfavorable for the freeze-in of SnV complexes. In addition, compared with the RTA terminal temperature of 1080 °C, the RTA terminal temperature of 1050 °C provides the free vacancies with more sufficient time to diffuse toward the sample surface inside the RTA furnace. Based on the above analysis, it can be understood why the total vacancy concentration in the bulk of SCZ silicon sample is also higher with the RTA terminal temperature of 1080 °C than with the RTA terminal temperature of 1050 °C. This explains why $[Au_s]$ shown in Fig. 7(b) reaches a maximum at the RTA terminal temperature of 1080 °C. As the RTA terminal temperature is 1020 °C, $[Au_s]$ continues to be a little smaller than that at the terminal temperature of 1080 °C, since the SnV complexes formed in the former case are somewhat fewer than those in the latter case, as shown in Fig. 6(b).

As the RTA terminal temperature decreases below 1020 °C, the evolution of vacancy-related defects enters a different regime. On the one hand, the longer cooling time inside the RTA furnace allows the free vacancies to continue diffusing toward the surface, leading to a further reduction in the total vacancy concentration retained in the bulk of the sample. On the other hand, a portion of the SnV complexes formed at higher temperatures can interact with the O_i atoms to form SnVO complexes within this temperature range. According to the previous MD results, the formation of SnVO complexes can begin at a certain temperature between approximately 1030 and 1020 °C. In addition, the theoretical calculations indicate that the SnVO complexes are relatively stable during the subsequent cooling process and are unlikely to

participate in the trapping of diffusing Au_i atoms to form Au_s atoms. Although the direct interaction kinetics between SnVO complexes and Au_i atoms have not been experimentally investigated in the present work, the DFT calculations suggest that the vacancy constituent in the SnVO complexes is strongly stabilized through the Sn–V–O binding configuration, thereby substantially reducing its availability for capturing diffusing Au_i atoms. Therefore, once the SnV complexes are transformed into the SnVO complexes, their contribution to the measured $[Au_s]$ signal in the DLTS spectra is expected to become negligible. Consequently, as the RTA terminal temperature is decreased from 1020 to 800 °C, the $[Au_s]$ continues to decrease. This trend reflects not only the progressive loss of free vacancies through the out-diffusion but also the gradual conversion of SnV complexes into SnVO complexes, which effectively removes the vacancy-related species from the population detectable by DLTS following the Au diffusion. It is worth noting that, as the RTA terminal temperature is decreased to 800 °C, the $[Au_s]$ values measured in the CZ and SCZ silicon wafers become comparable. This behavior can be understood by considering that, at such low terminal temperature, a substantial fraction of the SnV complexes in the SCZ silicon wafer has already been converted into SnVO complexes, thereby nearly removing the difference in the effective vacancy concentrations manifested by $[Au_s]$ values between the two types of silicon samples.

Collectively, based on the combined theoretical and experimental results described above, we establish a continuous, temperature-dependent sequence for vacancy-related defect evolution in SCZ silicon. Specifically, substantial SnV complex formation initiates within the temperature range of approximately 1100–1080 °C, where the vacancy capture by substitutional Sn atoms becomes predominant over the vacancy loss through out-diffusion. As the temperature further decreases to approximately 1030–1020 °C and below, part of the previously formed SnV complexes are progressively transformed into SnVO complexes through interactions with the O_i atoms. It should be emphasized that, although the cooling rate in our RTA experiments is much higher than that in actual SCZ silicon crystal growth, the governing mechanisms responsible for the formation and transformation of SnV and SnVO complexes are controlled primarily by the local defect reactions rather than by the absolute cooling rate itself. In practical crystal growth, the slower cooling rate provides longer dwell times within the aforementioned temperature windows, thereby facilitating more complete interactions among Sn atoms, vacancies, and O_i atoms and enabling the same sequence of defect evolution to occur more sufficiently. Therefore, the results presented above do not merely describe the isolated defect reactions, but instead define a physically consistent, temperature-resolved framework in which Sn-doping modifies the vacancy evolution pathways by redistributing vacancy-related species across distinct temperature windows during the cooling process of SCZ silicon crystal growth.

2. Effect of Sn-doping on the formation of grown-in oxide precipitates

The left panel of Fig. 8(a) plots the decrease in interstitial oxygen concentration $[\Delta(O_i)]$ as a function of the 1000 °C

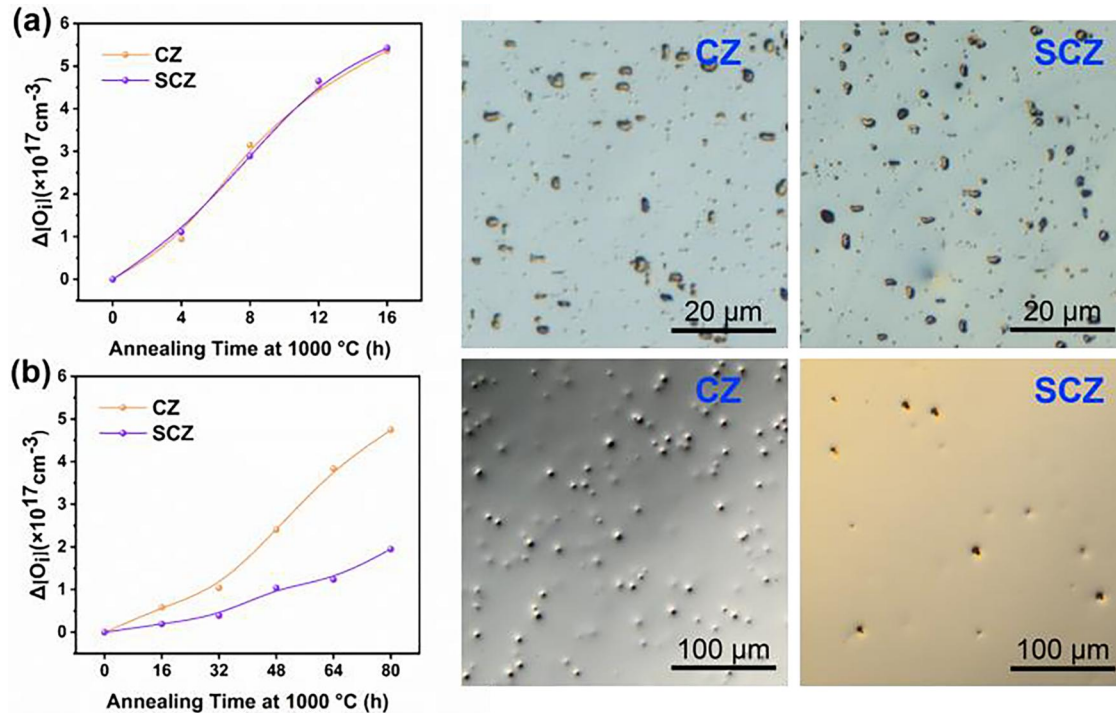


FIG. 8. (a) Left: Variation of $\Delta[O_i]$ during the subsequent low-high two-step ($800^\circ\text{C}/4\text{ h} + 1000^\circ\text{C}/0\text{--}16\text{ h}$) annealing for the $1250^\circ\text{C}/2\text{ h}$ pre-annealed CZ or SCZ silicon sample, as a function of the 1000°C annealing time. Right: Optical micrographs of the CZ and SCZ silicon samples subjected to $1250^\circ\text{C}/2\text{ h} + 800^\circ\text{C}/4\text{ h} + 1000^\circ\text{C}/16\text{ h}$ annealing, followed by preferential etching. (b) Left: Variation of $\Delta[O_i]$ for the as-received CZ and SCZ silicon samples subjected to the isothermal annealing at 1000°C ramped from 800°C at $1^\circ\text{C}/\text{min}$, as a function of the isothermal annealing time. Right: Optical micrographs of the as-received CZ and SCZ silicon samples subjected to 80 h annealing at 1000°C ramped from 800°C at $1^\circ\text{C}/\text{min}$, followed by the preferential etching.

annealing time, for the $1250^\circ\text{C}/2\text{ h}$ pre-annealed CZ and SCZ silicon samples subjected to the subsequent two-step annealing ($800^\circ\text{C}/4\text{ h} + 1000^\circ\text{C}/0\text{--}16\text{ h}$). The $1250^\circ\text{C}/2\text{ h}$ pre-annealing has effectively dissolved the grown-in oxide precipitates in both samples. Therefore, the observed $\Delta[O_i]$ results solely from oxygen precipitation with the nucleation and growth predominantly occurring at 800 and 1000°C , respectively. Under identical annealing conditions, the $\Delta[O_i]$ values for the CZ and SCZ silicon samples are nearly identical. Consistently, the optical micrographs [the right panel of Fig. 8(a)] reveal no obvious difference in the morphology, size, or density of the bulk microdefects (BMDs) associated with oxide precipitates between the CZ and SCZ silicon samples after the full annealing sequence ($1250^\circ\text{C}/2\text{ h} + 800^\circ\text{C}/4\text{ h} + 1000^\circ\text{C}/16\text{ h}$) and preferential etching. Given that the covalent radius of Sn is larger than that of Si, the substitutional Sn atoms introduce localized compressive lattice stress. The results in Fig. 8(a), however, indicate that at the [Sn] level studied here, this localized stress does not significantly alter the critical sizes required for the oxide precipitate nucleation and growth. Furthermore, the comparable $\Delta[O_i]$ kinetics in CZ and SCZ silicon imply that the present Sn-doping does not substantially affect oxygen diffusion, which is consistent with our earlier finding.⁴⁰

The left panel of Fig. 8(b) shows the variation of $\Delta[O_i]$ for the as-received CZ or SCZ silicon samples subjected to the isothermal annealing for up to 80 h at 1000°C ramped from 800°C at $1^\circ\text{C}/\text{min}$. According to the model of Kissinger *et al.* during the $1^\circ\text{C}/\text{min}$ ramp from 800 to 1000°C , the grown-in oxide precipitates that exceed the critical sizes for the growth within this temperature range will coarsen, whereas essentially no new oxide precipitate nuclei form.⁴¹ Therefore, the $\Delta[O_i]$ values shown in Fig. 8(b) (left) primarily arise from the growth of these grown-in oxide precipitates. Notably, $\Delta[O_i]$ is consistently and significantly lower in the SCZ silicon wafer than in the CZ counterpart. The right panel of Fig. 8(b) shows the optical micrographs of BMDs associated with the oxide precipitates formed in the as-received CZ and SCZ silicon samples after the full 80 h isothermal annealing at 1000°C ramped from 800°C , followed by the preferential etching. It is evident that the BMD density in the SCZ silicon sample is much lower than that in the CZ counterpart, indicating a lower density of grown-in oxide precipitates in the SCZ silicon. As established from Fig. 8(a), Sn-doping does not significantly affect the critical size required for the oxide precipitate growth during such thermal treatments. Therefore, we conclude that Sn-doping suppresses the formation of grown-in oxide precipitates during SCZ silicon crystal growth.

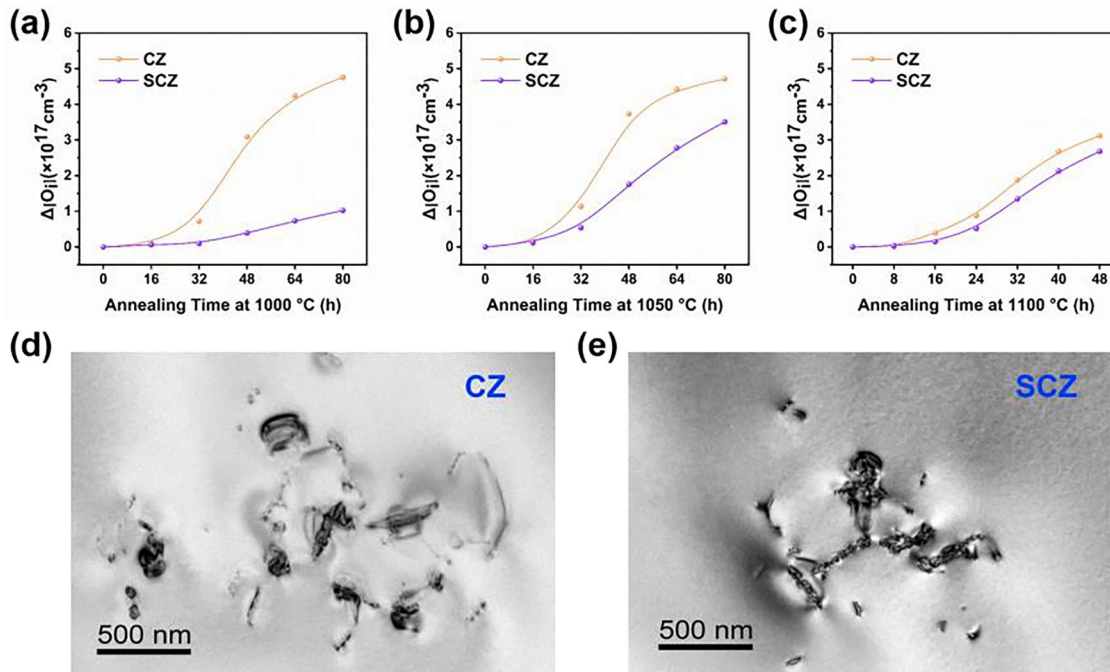


FIG. 9. (a)–(c) Dependences of $\Delta[O_i]$ on the isothermal annealing time at 1000, 1050, and 1100 °C, respectively, for the as-received CZ and SCZ silicon samples; (d) and (e) TEM images of the oxide precipitates and their induced defects formed in the CZ and SCZ silicon samples subjected to 1000 °C/80 h annealing.

Furthermore, Figs. 9(a)–9(c) plot the evolution of $\Delta[O_i]$ during the isothermal annealing at 1000, 1050, and 1100 °C, respectively, for the as-received CZ and SCZ silicon samples. According to the classical nucleation theory, during such isothermal annealing, only the grown-in oxide precipitates that exceed the critical size required for the growth at the isothermal annealing temperature can further coarsen. A key observation from Figs. 9(a)–9(c) is that $\Delta[O_i]$ is consistently higher in CZ silicon than in SCZ silicon across all annealing temperatures. Two plausible reasons may account for this: first, Sn-doping increases the critical size required for the oxide precipitate growth; second, SCZ silicon contains fewer grown-in oxide precipitates that exceed the critical size at the isothermal annealing temperature. The results in Fig. 8(a) allow us to preliminarily exclude the first reason. Nevertheless, to decisively identify the correct reason, we performed TEM observations on the representative samples annealed at 1000 °C for 80 h. We note that acquiring a representative TEM image was far more straightforward for CZ silicon [Fig. 9(d)] than for SCZ silicon [Fig. 9(e)], which stems from the significantly lower density of oxide precipitates in SCZ silicon. As shown in Figs. 9(d) and 9(e), the oxide precipitates in both CZ and SCZ silicon exhibit similar individual and aggregated forms, with no distinct differences in morphology and size. It is well known that the oxide precipitate growth is limited by oxygen diffusion. As established earlier, Sn-doping negligibly affects oxygen diffusion, and the associated lattice compressive stress does not significantly influence the oxide precipitate growth. Therefore, the comparable final

precipitate sizes (after 1000 °C/80 h annealing) imply that the initial sizes of the grown-in precipitates capable of coarsening at 1000 °C are also similar in both CZ and SCZ silicon. In other words, Sn-doping hardly affects the critical size for the growth of grown-in oxide precipitates. Thus, we can reasonably exclude the first possible reason. Consequently, the significantly weaker oxygen precipitation in SCZ silicon [Figs. 9(a)–9(c)] can be desirably attributed to a lower density of grown-in oxide precipitates. This result further confirms that Sn-doping suppresses the formation of grown-in oxide precipitates during SCZ silicon crystal growth.

We also indirectly compared the formation of grown-in oxide precipitates in CZ and SCZ silicon samples by investigating copper (Cu) precipitation. Figures 10(a) and 10(b) show the optical micrographs of the 1250 °C/2 h pre-annealed CZ and SCZ silicon samples, respectively, subjected to 1000 °C/5 min Cu diffusion, followed by the preferential etching. The etching reveals high-density dot-like pits in both samples, characteristic of homogeneously nucleated Cu precipitates as reported previously.⁴² Figure 10(c) shows a representative TEM image of the Cu precipitates in the aforementioned CZ sample. The precipitates are spherical particles 10–50 nm in diameter, with no obvious stress fields, dislocations, or other secondary defects in the surrounding matrix, consistent with the reported morphology of homogeneously nucleated Cu precipitates.⁴³ As mentioned earlier, the 1250 °C/2 h pre-annealing sufficiently eliminates the grown-in oxide precipitates. Therefore, upon cooling after Cu diffusion, the supersaturated Cu impurities undergo precipitation almost exclusively via homogeneous

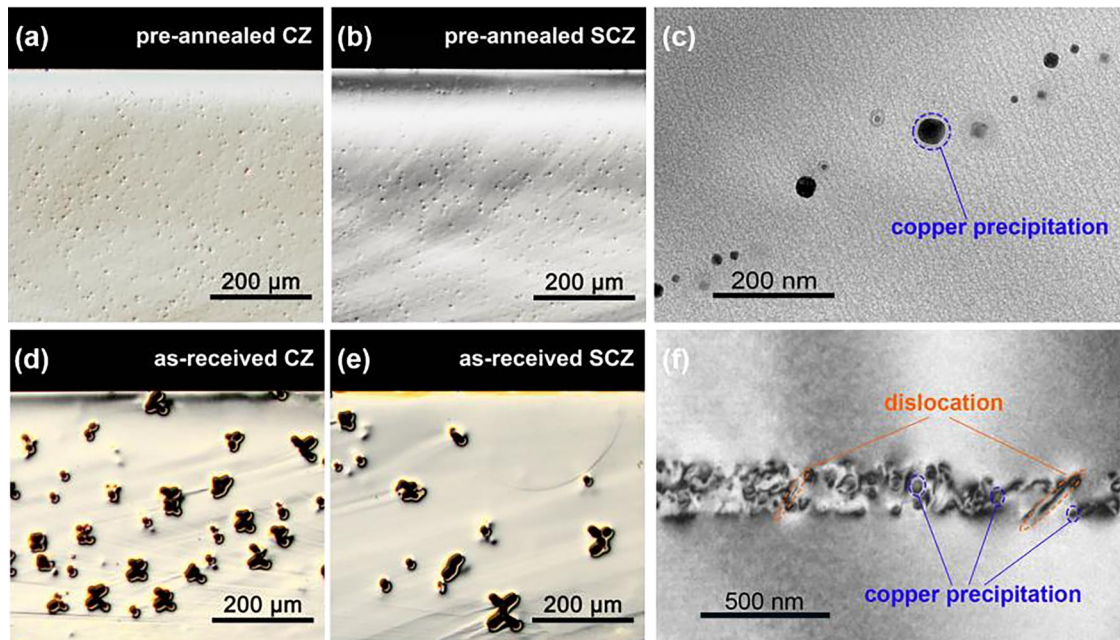


FIG. 10. (a) and (b) Optical micrographs for the 1250 °C/2 h pre-annealed CZ and SCZ silicon samples subjected to 1000 °C/5 min copper diffusion, followed by the preferential etching; (c) representative TEM image of the copper precipitates in the CZ silicon sample of (a); (d) and (e) optical micrographs for the as-received CZ and SCZ silicon samples subjected to the 1000 °C/5 min copper diffusion, followed by the preferential etching; (f) representative TEM image of the copper precipitates in the CZ silicon sample of (d).

nucleation. Notably, Figs. 10(a) and 10(b) reveal no obvious difference in Cu precipitate density between the two samples, indicating that Sn-doping itself hardly affects Cu precipitation. Figures 10(d) and 10(e) show the optical micrographs of the as-received CZ and SCZ silicon samples, respectively, subjected to 1000 °C/5 min Cu diffusion, followed by the preferential etching. The etching morphologies in both samples are similar, appearing as aggregated rod-like patterns, which is distinctly different from the dot-like pits in Figs. 10(a) and 10(b). Furthermore, the Cu precipitate density in the SCZ silicon sample is significantly lower than that in the CZ wafer. Figure 10(f) shows a representative TEM image of the Cu precipitates in the CZ silicon sample of Fig. 10(d). The Cu precipitates are agglomerates composed of numerous 10–20 nm particles. Significant stress and dislocations are evident around these agglomerates, consistent with the reported characteristics of the heterogeneously nucleated Cu precipitates.⁴⁴ The as-received silicon wafers contain grown-in oxide precipitates, which can serve as the heterogeneous nucleation sites for Cu precipitation. Clearly, the heterogeneous nucleation on these sites has a significantly lower energy barrier than the homogeneous nucleation. Therefore, the supersaturated Cu impurities preferentially nucleate heterogeneously on the grown-in oxide precipitates upon cooling after Cu diffusion. The growth of these Cu precipitates generates secondary defects such as dislocations, which then act as additional heterogeneous sites,⁴⁵ leading to the observed aggregation of Cu precipitates. In this context, the homogeneous

nucleation for Cu precipitation is substantially suppressed. Thus, the Cu precipitates in Figs. 10(d) and 10(e) are formed predominantly via the heterogeneous nucleation on the grown-in oxide precipitates. Consequently, the significantly lower Cu precipitate density in the SCZ silicon sample indirectly indicates that Sn-doping suppresses the formation of grown-in oxide precipitates.

3. Effect of Sn-doping on void formation

Figures 11(a) and 11(b) show the representative optical micrographs of the as-received CZ and SCZ silicon samples after the vertical immersion in Secco etchant for 30 min preferential etching. The wedge-shaped etching patterns, corresponding to the FPDs, are observed on both samples, with a visibly lower density in the SCZ silicon sample. Yamagishi *et al.* first confirmed that the FPDs are related to vacancy-type D-defects.⁴⁶ Generally, the D-defects manifest as the voids formed by the vacancy aggregation in the CZ-grown silicon crystals.^{47,48} Secco etching is, therefore, a rapid method for revealing voids via FPDs, especially suitable for laboratory research activities. For quantification, the FPDs were imaged at 20 locations on each wafer under identical conditions, and their area densities were calculated. The average FPD densities and standard deviations are summarized in Fig. 11(c). The average density is approximately 20% lower in SCZ silicon than in CZ silicon, indicating that Sn-doping suppresses void formation during SCZ silicon crystal growth.

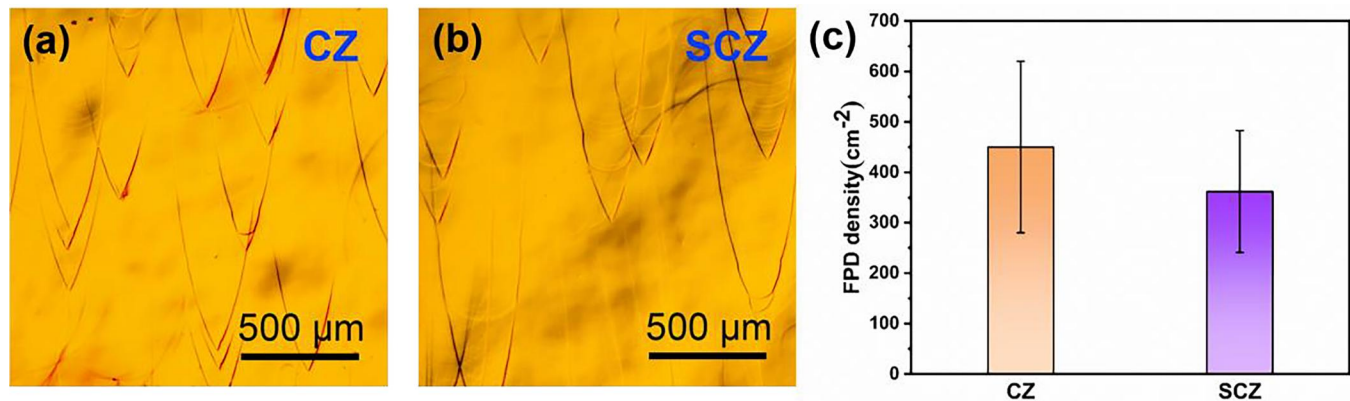


FIG. 11. (a) and (b) Representative optical micrographs of the FPDs delineated by Secco preferential etching in the as-received CZ and SCZ silicon samples; (c) comparison of FPD area densities in the as-received CZ and SCZ silicon samples.

C. On the physical mechanism of Sn-doping effects on grown-in defect formation

As described above, we have experimentally confirmed that Sn-doping suppresses the formation of grown-in oxide precipitates and voids in CZ-grown silicon. Meanwhile, through DFT calculations, MD simulations, and tailored experiments including RTA, gold diffusion, and DLTS, we have elucidated the effects of Sn-doping on point defect formation and on the evolution pathways of SnV and SnVO complexes. Based on the experimental and theoretical results obtained in this work, together with previous reports,^{3,34} the physical mechanism underlying the effects of Sn-doping on grown-in defect formation in CZ-grown silicon can be described as follows.

Under vacancy-rich CZ growth conditions, as the crystal gradually cools at temperatures above approximately 1100 °C, the excess vacancies remaining after the recombination of vacancies and self-interstitial silicon atoms exist primarily as free vacancies. Subsequently, the evolution of vacancy-related defects is governed by the competition between vacancy aggregation, vacancy trapping by Sn atoms, and vacancy-oxygen pairing. At elevated temperatures relevant to void formation, substantial SnV complex formation initiates in the range of 1100–1080 °C. Under these conditions, the interaction between vacancies and substitutional Sn atoms proceeds via a trap-limited mechanism, in which highly mobile vacancies are efficiently captured by essentially immobile Sn atoms to form SnV complexes. Importantly, this temperature regime largely overlaps with the void-forming regime. Consequently, SnV complex formation and void formation occur concurrently and compete for the same population of mobile free vacancies. The continuous trapping of vacancies into SnV complexes progressively reduces the vacancy population available for aggregation, thereby suppressing void nucleation and growth. As the temperature further decreases to below ~1030 °C, the system enters a regime in which V–O interactions become kinetically accessible. In conventional CZ silicon, this leads to the formation of VO₂ complexes that act as effective heterogeneous nucleation centers for oxygen precipitation. In SCZ silicon, however, a

substantial fraction of vacancies has already been incorporated into SnV complexes at higher temperatures. These SnV complexes subsequently react with O_i atoms to form SnVO complexes, which exhibit negligible binding affinity for additional oxygen atoms and, therefore, cannot evolve into oxygen precipitation nuclei. Meanwhile, the reduced population of mobile free vacancies further suppresses the formation of conventional VO₂ complexes. Collectively, these results reveal a thermodynamic-kinetic decoupling in vacancy behavior induced by Sn-doping: although Sn-doping lowers the vacancy formation energy, it simultaneously reduces the population of mobile free vacancies through efficient trapping into relatively immobile SnV complexes. As a consequence, both void formation and oxygen precipitation are suppressed not by reducing the overall vacancy population but by redistributing vacancies into kinetically inactive configurations.

More broadly, this work demonstrates that defect engineering in CZ-grown silicon should be understood within a kinetic framework involving competition between vacancy trapping and vacancy aggregation pathways during crystal cooling. This competition governs the redistribution of mobile vacancies among distinct defect evolution pathways and thereby determines the formation of vacancy-related defects. The present framework provides a mechanistic foundation for understanding and designing co-doping strategies for impurity-mediated defect engineering in silicon crystals.

V. CONCLUSIONS

In this work, the effects of Sn-doping on vacancy-related defect evolution in Czochralski silicon have been systematically investigated through the combination of first-principles calculations, molecular dynamics simulations, rapid thermal annealing, Au diffusion, and DLTS characterization. The results demonstrate that Sn-doping fundamentally modifies the kinetic evolution pathways of vacancy-related defects during crystal cooling under low intrinsic vacancy concentrations relevant to practical CZ crystal growth.

The theoretical calculations and experimental results consistently show that substantial SnV complex formation initiates within the temperature range of 1100–1080 °C and continues down to ~1030 °C. Since this temperature window substantially overlaps with the regime for vacancy aggregation and void evolution, the trapping of mobile vacancies by substitutional Sn atoms kinetically competes with void nucleation and growth during cooling. As the temperature further decreases below ~1030 °C, part of the SnV complexes evolve into relatively stable SnVO complexes through interaction with interstitial oxygen atoms. Unlike conventional VO-related complexes, the SnVO complexes exhibit negligible binding affinity for additional oxygen atoms and, therefore, cannot act as effective heterogeneous nucleation centers for oxygen precipitation. Meanwhile, the continuous trapping and redistribution of mobile vacancies further suppress the formation of VO₂ complexes. Consequently, the formation of both voids and grown-in oxide precipitates is significantly suppressed in SCZ silicon.

More importantly, this work demonstrates that the evolution of vacancy-related defects during crystal cooling cannot be understood solely from defect energetics, but must be analyzed within a kinetic framework involving competing reaction pathways under non-equilibrium cooling conditions. The combined RTA, Au diffusion, and DLTS methodology developed in this work further provides a generally applicable experimental platform for probing impurity-mediated vacancy evolution under low vacancy concentrations.

It is worth noting that similar defect-suppression behavior is also observed at lower Sn-doping concentrations (e.g., $3 \times 10^{17} \text{ cm}^{-3}$), although the extent of suppression becomes weaker because of the reduced availability of vacancy trapping sites at lower Sn concentrations. Unlike previous studies on electron-irradiated Sn-doped silicon containing considerably high vacancy concentrations, the present work focuses on Sn–V–O interactions under the much lower intrinsic vacancy concentrations relevant to practical crystal growth conditions, thereby providing a physically realistic description of defect evolution during crystal cooling. Overall, the present results demonstrate that isovalent Sn-doping provides an effective means of controlling vacancy-mediated defect evolution in CZ silicon through kinetically driven vacancy redistribution across distinct temperature regimes during crystal cooling. In this sense, SCZ silicon represents a promising material platform for fabricating devices requiring low densities of grown-in oxide precipitates and void-related defects. This is particularly relevant to power devices such as insulated-gate bipolar transistors (IGBTs), for which oxide precipitates and their associated secondary defects may act as carrier recombination centers and increase leakage current, whereas voids can compromise gate oxide integrity (GOI). Therefore, by simultaneously suppressing grown-in oxide precipitates and voids, SCZ silicon wafers may offer potential advantages in reducing leakage current and improving GOI, warranting future device-level validation.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Qunlin Nie: Data curation (lead); Formal analysis (equal); Methodology (equal); Validation (lead); Visualization (equal); Writing – original draft (lead). **Hao Chen:** Data curation (supporting); Resources (equal); Validation (equal). **Defan Wu:** Data curation (supporting); Validation (equal). **Ruokai Wu:** Data curation (supporting); Resources (equal). **Anchen Tang:** Data curation (supporting); Resources (equal). **Tong Zhao:** Formal analysis (lead); Methodology (lead); Resources (equal); Supervision (lead); Visualization (equal); Writing – original draft (lead). **Deren Yang:** Resources (equal); Supervision (equal). **Xiangyang Ma:** Conceptualization (lead); Resources (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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